project proposal

1. Set up

Packages

```r
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# read .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# arrange plots
library(patchwork)
```

Data

```r
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
```

```r
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")


# read in site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# read in water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

2. Cleaning

`sites` object

```r
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

Cleaning `taxon_list`

```r
# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))
```

Cleaning `water_quality`

```r
# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
```

```
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(med_ph = median(dissolved_oxygen_mg_l, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")
```

Cleaning `aq_ins`

```
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
```

```r
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
               values_to = "abundance") |>
  # group by date, site, taxon
  group_by(date, site, taxon) |>
  # calculate average abundance per liter (sum abundance divided by 7.5 liters)
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
  # ungroup data frame
  ungroup() |>
  # create a new column called `date_site` to join with water quality
  unite("date_site", date, site, remove = FALSE) |>
  # join with water quality
  left_join(water_quality_clean, by = c("date_site")) |>
  # join with taxon list
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  # add full names of sites
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>
  # setting factor levels for site
  mutate(site_full = fct_reorder(site_full, med_do, .fun = "median", .na_rm = TRUE),
         site_full = fct_rev(site_full))
```

Focus Taxa

```r
# taxa to include in all figures
focus_taxa <- c("Ostracod", "Copepod", "Cladocera",
                "Hemiptera Corixidae Boatman", "Nematode")
```

Figure 1: Invertebrate abundance as a function of DO

```r
# create object sal_taxa from aq_ins_clean
DO_taxa <- aq_ins_clean |>
  # remove rows that are missing median salinity values
  filter(!is.na(med_do)) |>
  # create column that calculates log-transformed abundance
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # mutates to title-case taxon names
  mutate(taxon = str_replace_all(taxon, "_", " "), taxon = str_to_title(taxon)) |>
  # only include focus taxa
  filter(taxon %in% focus_taxa) |>
  # order taxon levels by maximum average abundance per liter
  mutate(taxon = fct_reorder(taxon, ave_lit, .fun = max, .desc = TRUE)) |>
  # keep only the columns needed for the figure
  select(date, site, site_full, taxon, ave_lit, log_ave_lit, med_do)
```

```r
# base layer: ggplot
DO_taxa_plot <- ggplot(data = DO_taxa, # assigns plot to object do_taxa_plot
                        # uses sal_taxa data frame
                        mapping = aes(x = med_do, # x-axis: med_do
                                      y = log_ave_lit, # y-axis: log_ave_lit
                                      color = taxon)) + # color by taxon
  # first layer: points show individual abundance measurements
  geom_point(shape = 16, # shape is circle
             size = 2, # size is 2
             alpha = 0.7) + # opacity is 0.7
  # separate panels for each taxon with free x- and y-axis scales
  facet_wrap(~ taxon, scales = "free") +
  # use non-default colors for points
  scale_color_viridis_d() +
  labs(x = "Median DO (mg/L)", # relabel x-axis
       y = "log + 1 transformed average abundance/L", # relabel y-axis
       # creates descriptive title
       title = "Invertebrate abundance as a function of DO") +
  # uses non-default theme
  theme_minimal() +
  # remove redundant legend
  theme(legend.position = "none")

# shows plot
```

```
DO_taxa_plot
```

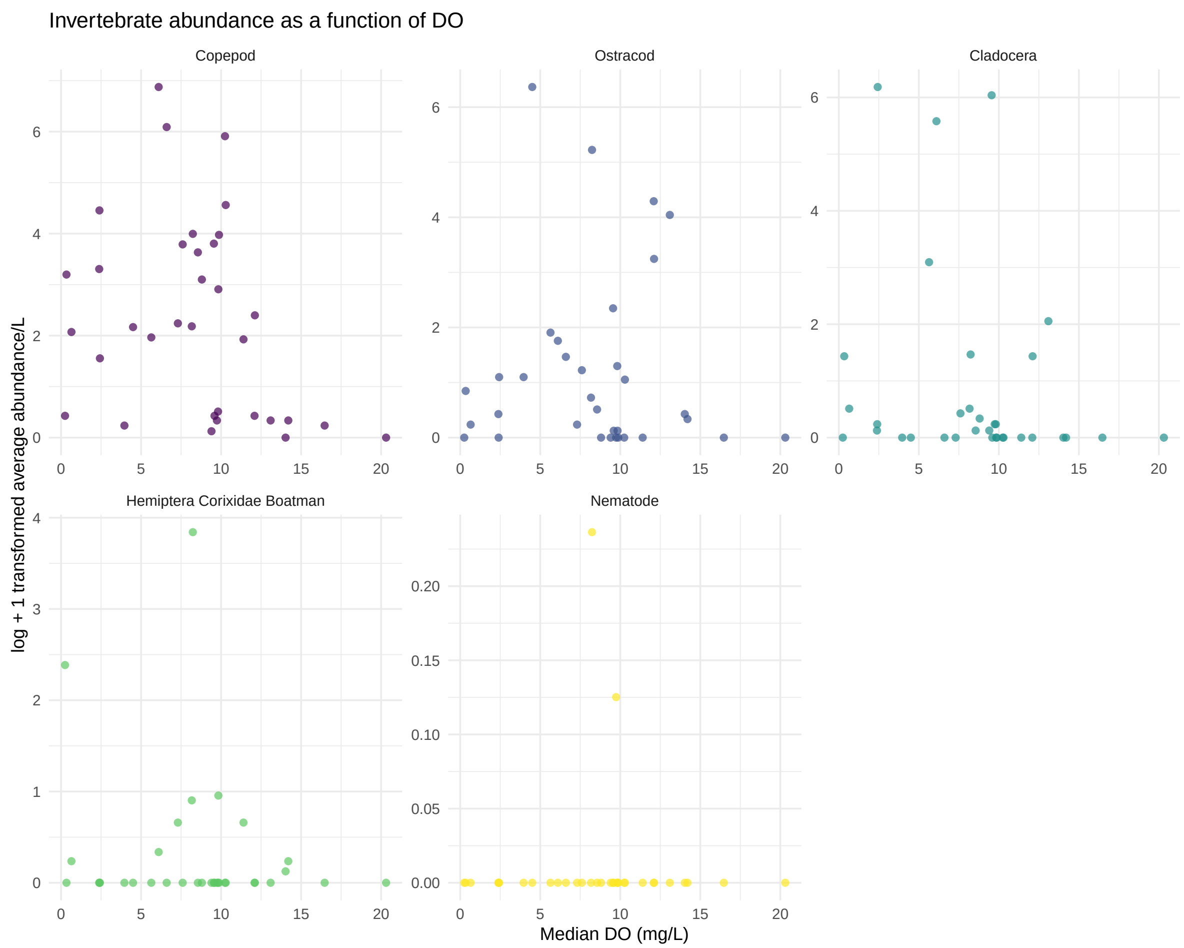


Figure 2: Variance of DO across sights

```
# create object aq_ins_clean_fig2 from aq_ins_clean
aq_ins_clean_fig2 <-  aq_ins_clean |>
  # convert taxon to title case
  mutate(taxon = str_replace_all(taxon, "_", " "),
         taxon = str_to_title(taxon)) |>
  # filter to include focus taxa
  filter(taxon %in% focus_taxa)
```

```r
# base layer
DO <- ggplot(data = aq_ins_clean_fig2,
       # x-axis: site
       # y-axis: mean salinity
       mapping = aes(x = site_full,
                     y = med_do)) +
  # first layer: boxplot
  geom_boxplot(width = 0.35,
               outlier.shape = NA) +
  # second layer: salinity measurements at each survey
  geom_jitter(height = 0,
              width = 0.12,
              shape = 21,
              alpha = 0.6) +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(width = 15)) +
  # second layer: points to represent means
  stat_summary(fun = median,
               color = "red",
               size = 1) +
  # relabelling x- and y-axes
  labs(x = "Site",
       y = "Median DO (mh/L)",
       title = "Site DO levels") +
  # different theme
  theme_bw()
# show plot
DO
```

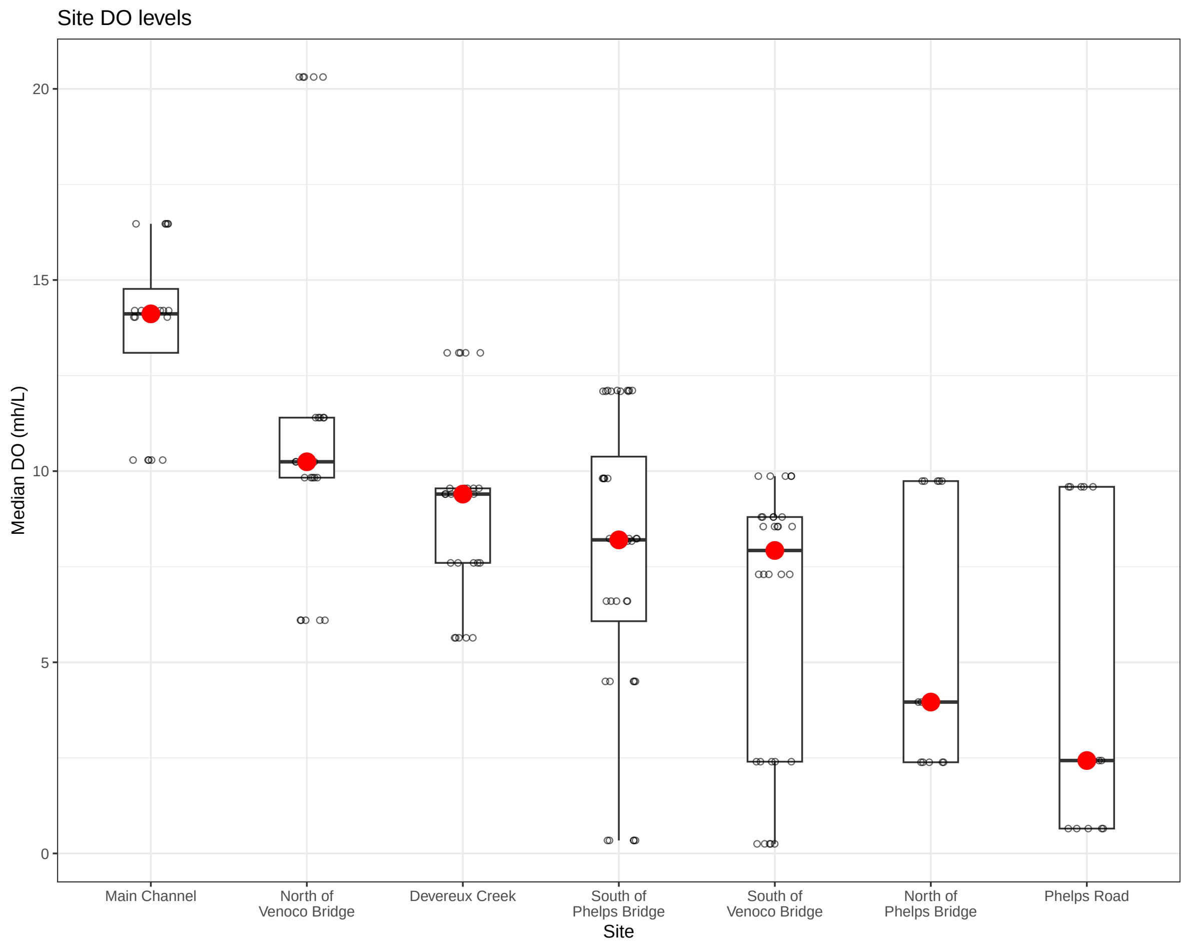


Figure 3: Aquatic Invert. Abundance Across Sites

```
# creating new object from clean aquatic inverts
abundance_by_site <- aq_ins_clean |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # convert taxon to title case
  mutate(taxon = str_replace_all(taxon, "_", " "),
         taxon = str_to_title(taxon)) |>
  # filter to include taxa of interest
  filter(taxon %in% focus_taxa)
```

```r
# base layer

abundance <- ggplot(data = abundance_by_site,
                    # x-axis: site
                    # y-axis: mean abundance/L
                    # color by taxon
                    mapping = aes(x = site_full,
                                  y = log_ave_lit,
                                  color = taxon)) +
  # first layer: jitter points to represent observations
  geom_jitter(width = 0.15,
              height = 0,
              alpha = 0.7) +
  # facet by taxon
  facet_wrap(~ taxon) +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # non-default colors
  scale_color_viridis_d() +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Aquatic Invert. Abundance Across Sites") +
  theme_bw() +
  # remove legend
  theme(legend.position = 'none',
        # fix x-axis for readability
        axis.text.x = element_text(angle = 45, hjust = 1))

abundance
```

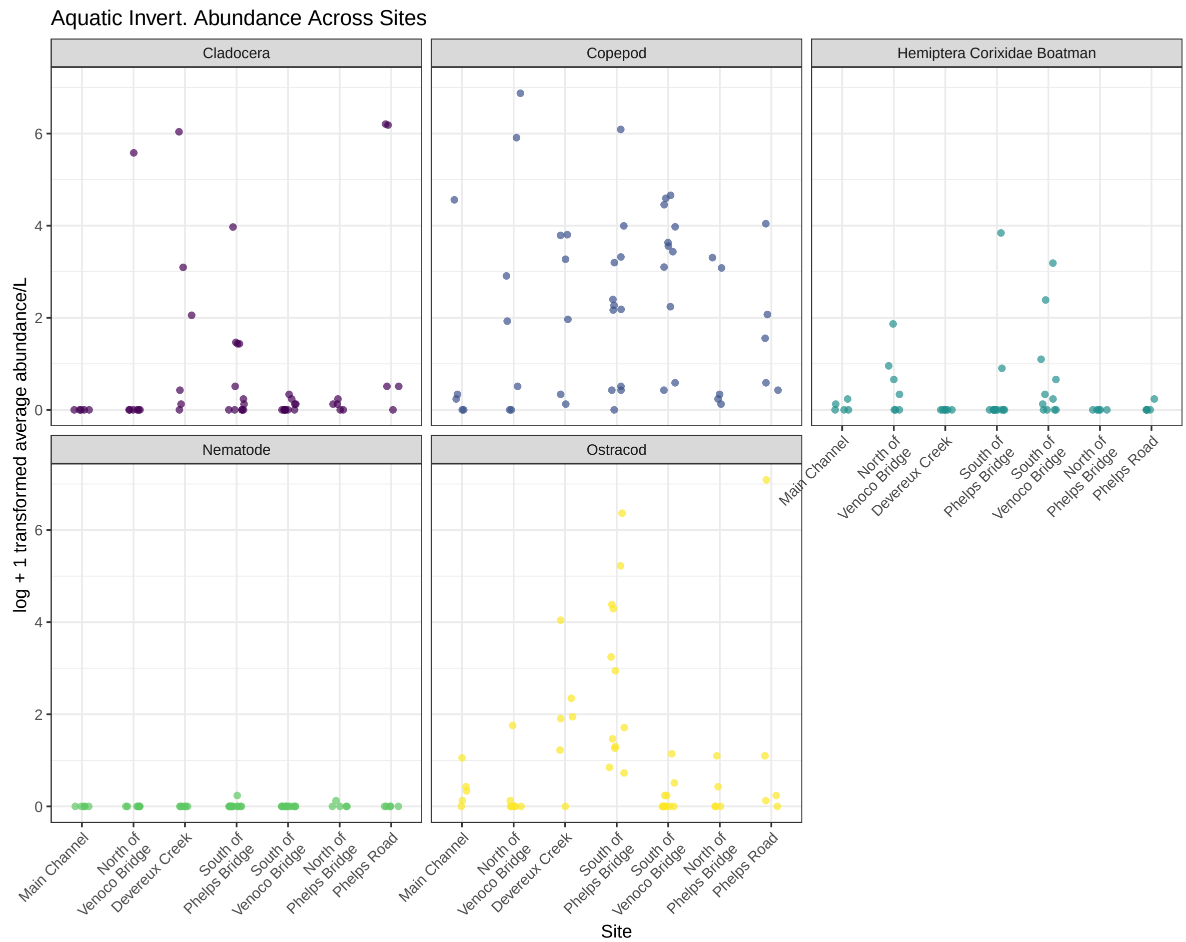


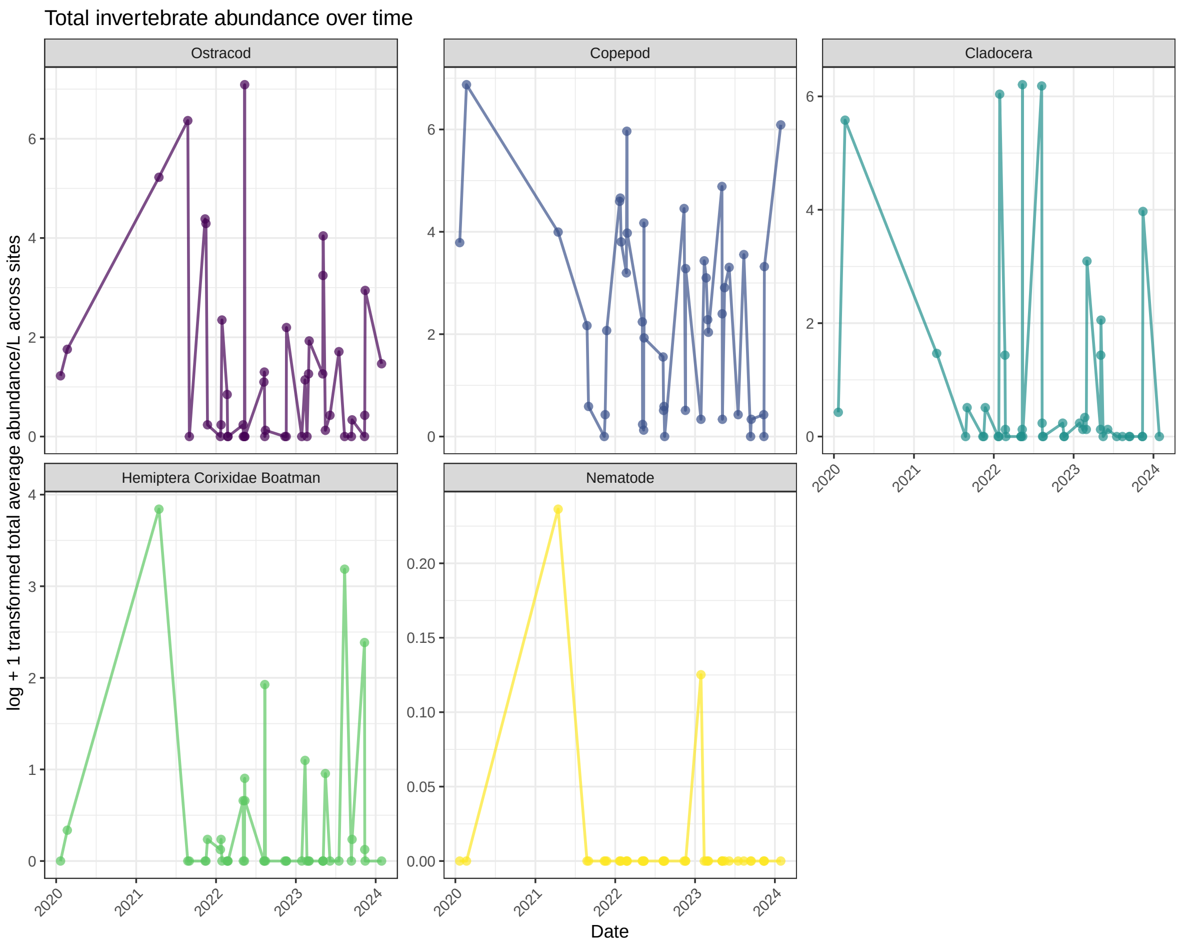
Figure 4: Abundance over time

```
# create object for plotting total abundance over time
abundance_time <- aq_ins_clean |>
  # group by sampling date and taxon
  group_by(date, taxon) |>
  # total abundance across all sites for each date and taxon
  summarize(total_ave_lit = sum(ave_lit, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # create log-transformed total abundance column
  mutate(log_total_ave_lit = log(total_ave_lit + 1)) |>
  # make taxon names easier to read
```

```r
  mutate(taxon = str_replace_all(taxon, "_", " "),
         taxon = str_to_title(taxon)) |>
  # order taxa by maximum total abundance
  mutate(taxon = fct_reorder(taxon, total_ave_lit, .fun = max, .desc = TRUE)) |>
  # include main taxon groups of focus
  filter(taxon %in% c("Ostracod", "Copepod", 'Cladocera',
                      'Hemiptera Corixidae Boatman', 'Nematode'))

# base layer
abundance_time_plot <- ggplot(data = abundance_time,
                              # x-axis: sampling date
                              # y-axis: total log-transformed abundance
                              mapping = aes(x = date,
                                            y = log_total_ave_lit,
                                            color = taxon)) +
  # points show total abundance on each sampling date
  geom_point(size = 2,
             alpha = 0.7) +
  # lines connect total abundance through time for each taxon
  geom_line(linewidth = 0.8,
            alpha = 0.7) +
  # separate panels for each taxon
  facet_wrap(~ taxon, scales = "free_y") +
  # use non-default colors
  scale_color_viridis_d() +
  # relabel axes and add title
  labs(x = "Date",
       y = "log + 1 transformed total average abundance/L across sites",
       title = "Total invertebrate abundance over time") +
  # use clean theme
  theme_bw() +
  # remove redundant legend
  theme(legend.position = "none",
        axis.text.x = element_text(angle = 45,
                                   hjust = 1))

# show plot
abundance_time_plot
```


Total invertebrate abundance over time
Ostracod
Copepod
Cladocera
Hemiptera Corixidae Boatman
Nematode
log + 1 transformed total average abundance/L across sites
Date
2020
2021
2022
2023
2024
0
2
4
6
1
3
0.00
0.05
0.10
0.15
0.20